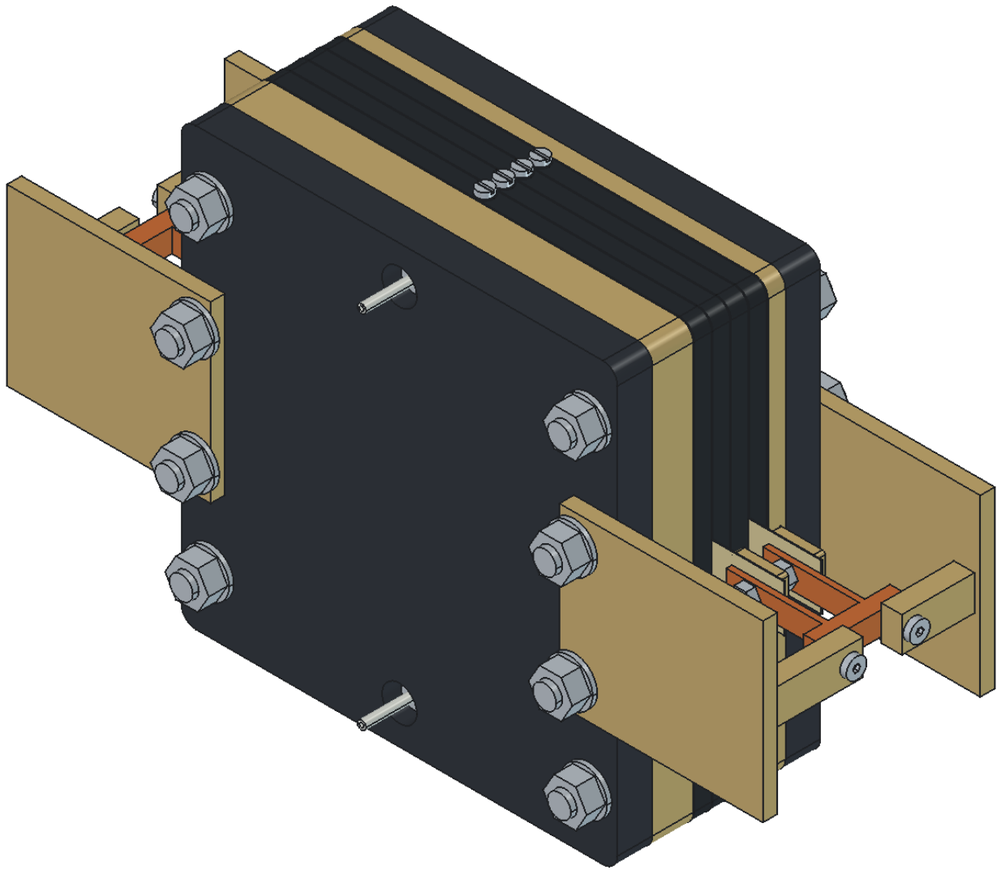
01 / 紧凑板框电堆 - 前端集中进出液



建议先打开五电极原生装配

assemblies/Kolbe_5.FCStd

可导入其他 CAD 的实体文件

assemblies/Kolbe_5.step

150 × 150 mm 主体板面

5 电极 / 4 并联流道

主体轴向厚度 69.75 mm

照片式板框夹紧，内部共用歧管

金属电极主体 56 × 56 mm

绝缘载片承担歧管与外围定位

所有 A 并联，所有 C 并联

铜梳排有独立绝缘支架

净距 6 mm 是评审尺寸。

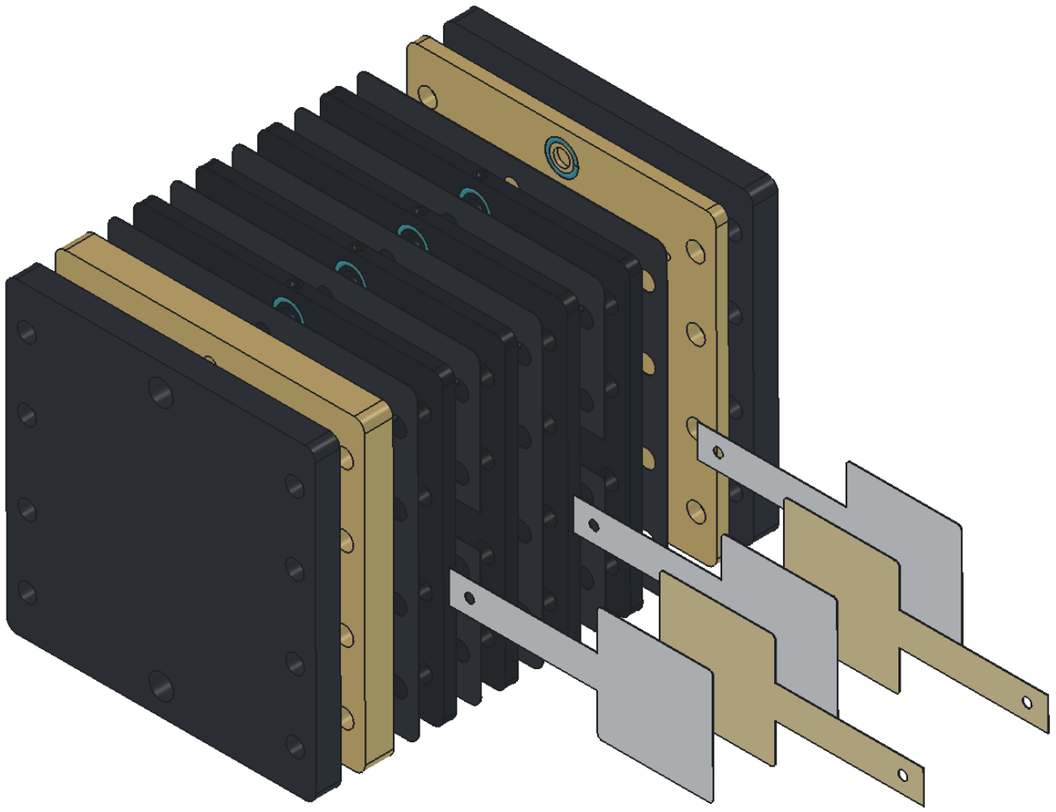
实际电极、泵流量与压力待提供。

密封材料、预紧与端板受力待核定。

图中外侧支架用于支撑干侧梳排。

本版未做压力、漏液或热态试验。

02 / 分层拆装 - 电极与绝缘载片分离



从前至后：

前压板 / 前绝缘液路板

C / 流道框 / A / 流道框 / C

流道框 / A / 流道框 / C

后绝缘封液板 / 后压板

每张金属片配同厚绝缘载片。

主密封圈仅压金属片窄边。

歧管密封圈压在绝缘载片上。

内部电极两面均面向流道。

最外侧阴极主要使用朝内一面。

爆炸图横向移出金属电极，

用于检查反应窗、耳片与载片。

铜连接与紧固件在本图中省略。

单件 FCStd 保留原生特征树。

装配零件定义嵌入同一文件，

无外部零件引用。

定位采用明确基准放置；

未添加运动学装配约束。

03 / 电极面积 - 窄边密封，外围由绝缘载片承担

金属主体 56 × 56 / R2；窗口 50 × 50 / R3



橙线：密封槽边界（中心线 53.6 / R4.8；宽 1.4）

外槽边至金属边名义余量 0.5 mm；需结合公差与薄片变形核查。

按圆角轮廓计算：

单面窗口面积：2492.27 mm ²

金属主体面积：3132.57 mm ²

主体几何面积占比：79.56%

计入当前干侧耳片后，金属总面积 4284.57 mm ²，

窗口占金属总面积 58.17%。

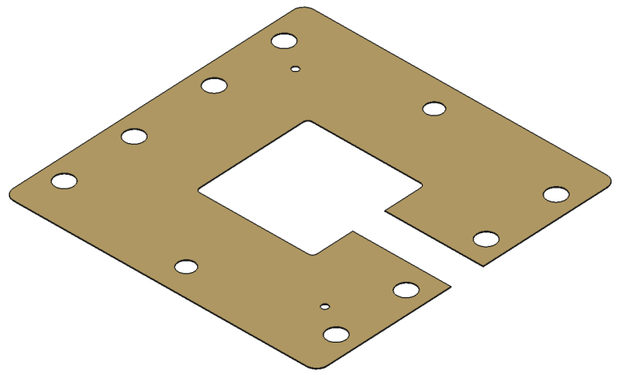
这是几何开窗面积，不是实测电化学有效面积。

载片不覆盖金属反应面。

实际电极尺寸未提供，主体和窗口均可重做。

如采用镀铂钛，可将铂镀层限制在工作区，

耳片与密封边的镀覆范围另行明确。



150 mm 绝缘载片承担歧管孔、定位与外围。

04 / 实际 CAD 核验与待定工程输入

电极数	内部流道	主体厚度 / mm	STEP 实体数	未解释实体干涉
3	2	57.05	103	0
5	4	69.75	135	0
7	6	82.45	167	0

已核验：原生装配可打开；全部实例为有效实体；STEP 重新导入实体数量一致。

螺纹采用名义包络。M4 螺钉与攻丝底孔的设计啮合已单列，不属于遗漏的碰撞。

密封件装配形状为保面积椭圆截面示意，供货尺寸以独立自由态零件为准。

仍需：实际电极长宽厚及材料、泵工作流量与最大压力、电解液/溶剂、温度和电流。

据此确定净距、支路孔径、材料、密封圈、公差、预紧以及端板挠度要求。

内部共用歧管仍需检查旁路电流、并联流量分配和气体排放路径。

未执行实际加工、压力/漏液、介质相容性、热循环和电气温升试验。

后附原生实体投影与剖视图。每个定制件的 FCStd 与 STEP 见 parts/。

KOLBE ELECTROCHEMICAL CELL / REV B1

板式压紧电化学电池

CAD尺寸评审图册

依据最终零件BREP生成正视、侧视、轴测投影；尺寸单位为mm。

湿窗投影占箔片实体：79.56% = 2492.274 / 3132.566 mm²（50 x 50 R3窗口 / 56 x 56 R2箔片本体；干端子排除）。

审查状态：压力、化学兼容性、材料牌号、制造公差及密封验证未完成。

6 mm间隙仅为机械评审包络；实际电极厚度、装配间隙和泵流量待确认。

不赋予额定压力或最终紧固扭矩。螺纹标注为名义光滑包络，不代表螺旋牙型。

图册包含零件功能、材料现状、主要尺寸、端口/密封/紧固特征和模型投影。

包含3/5/7层当前总成投影及全部在用自定义零件；已剔除未使用的旧端子、铜带和固定长度定位销。

图纸编号：B1-CAD-REVIEW | 禁止用于制造放行

KOLBE-3 | 3-LAYER PLATE-PRESS STACK

Actual assembled BREP projection | See component sheets for manufacturing dimensions

ASSEMBLY ENVELOPE / 总成参数

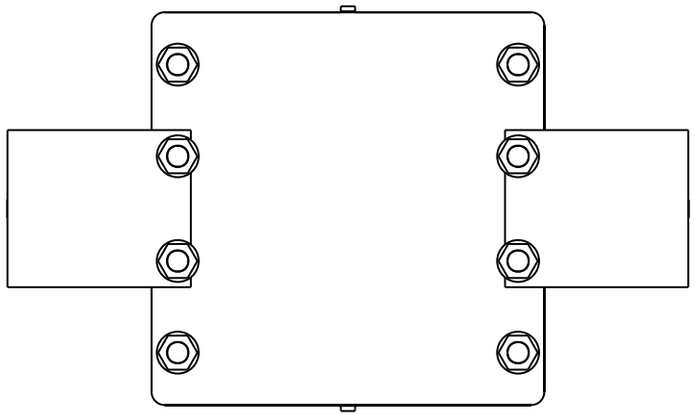
01 BREP envelope: 260.4 x 154.6 x 87.25 mm (X x Y x Z)

02 Validated body stack height: 57.05 mm; active layer-stack height: 13.05 mm

03 Assembly BREP contains 103 component instances; view shows actual model placements

04 Geometry/model volume reported by assembly Stack: 1363161.9 mm³; not a pressure or strength rating

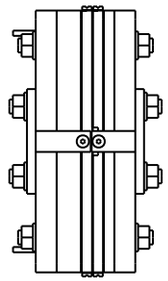
05 6 mm inter-electrode gap is a provisional envelope; pump flow, materials and sealing remain unresolved



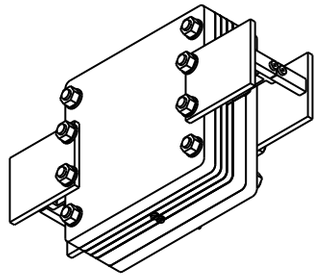
PLAN / FRONT (1:2.1)



SECTION A-A (Y=0; Z MID-STACK)



SIDE (1:3.1)



ISOMETRIC (1:4.2)

REVIEW GEOMETRY ONLY; do not infer pressure, torque, current capacity or final gap from this assembly view.

ASSEMBLY	DRAWING	SOURCE	STATUS
3-layer compact plate-press	**KOLBE-3 / ASM-3**	Kolbe_3.FCStd Stack.Shape	**REVIEW**

KOLBE-5 | 5-LAYER PLATE-PRESS STACK

Actual assembled BREP projection | See component sheets for manufacturing dimensions

ASSEMBLY ENVELOPE / 总成参数

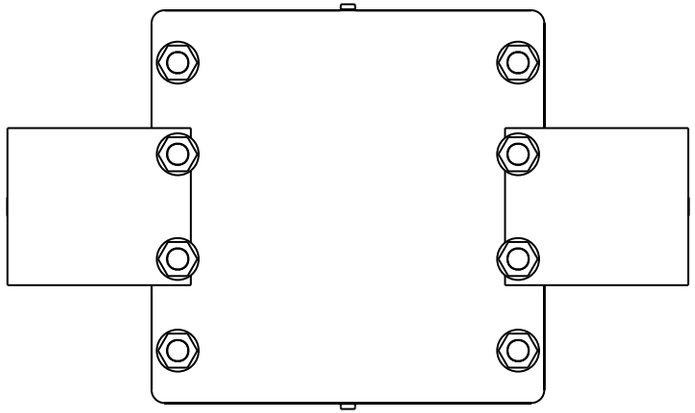
01 BREP envelope: 260.4 x 154.6 x 99.95 mm (X x Y x Z)

02 Validated body stack height: 69.75 mm; active layer-stack height: 25.75 mm

03 Assembly BREP contains 135 component instances; view shows actual model placements

04 Geometry/model volume reported by assembly Stack: 1619713.1 mm³; not a pressure or strength rating

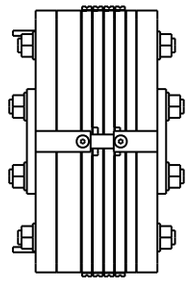
05 6 mm inter-electrode gap is a provisional envelope; pump flow, materials and sealing remain unresolved



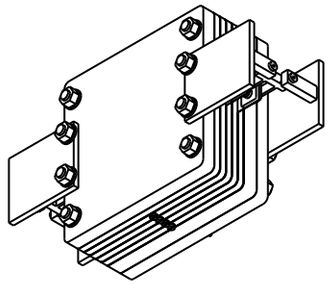
PLAN / FRONT (1:2.1)



SECTION A-A (Y=0; Z MID-STACK)



SIDE (1:3.1)



ISOMETRIC (1:4.2)

REVIEW GEOMETRY ONLY; do not infer pressure, torque, current capacity or final gap from this assembly view.

ASSEMBLY	DRAWING	SOURCE	STATUS
5-layer compact plate-press	**KOLBE-5 / ASM-5**	Kolbe_5.FCStd Stack.Shape	**REVIEW**

KOLBE-7 | 7-LAYER PLATE-PRESS STACK

Actual assembled BREP projection | See component sheets for manufacturing dimensions

ASSEMBLY ENVELOPE / 总成参数

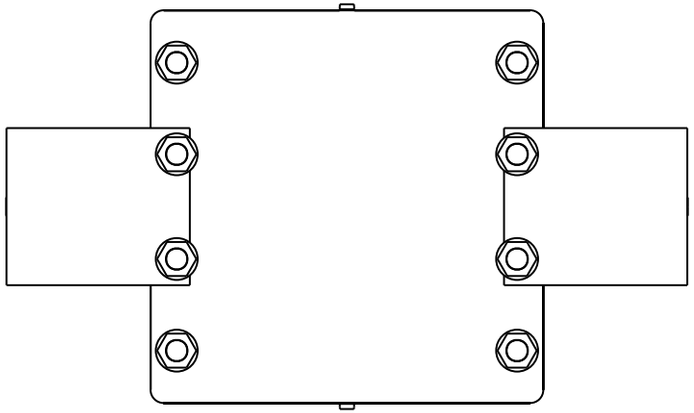
01 BREP envelope: 260.4 x 154.6 x 112.65 mm (X x Y x Z)

02 Validated body stack height: 82.45 mm; active layer-stack height: 38.45 mm

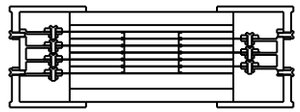
03 Assembly BREP contains 167 component instances; view shows actual model placements

04 Geometry/model volume reported by assembly Stack: 1876264.2 mm³; not a pressure or strength rating

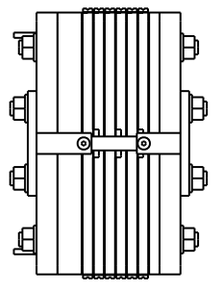
05 6 mm inter-electrode gap is a provisional envelope; pump flow, materials and sealing remain unresolved



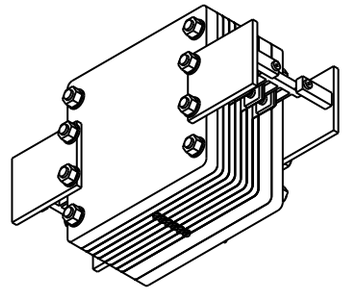
PLAN / FRONT (1:2.1)



SECTION A-A (Y=0; Z MID-STACK)



SIDE (1:3.1)



ISOMETRIC (1:4.2)

REVIEW GEOMETRY ONLY; do not infer pressure, torque, current capacity or final gap from this assembly view.

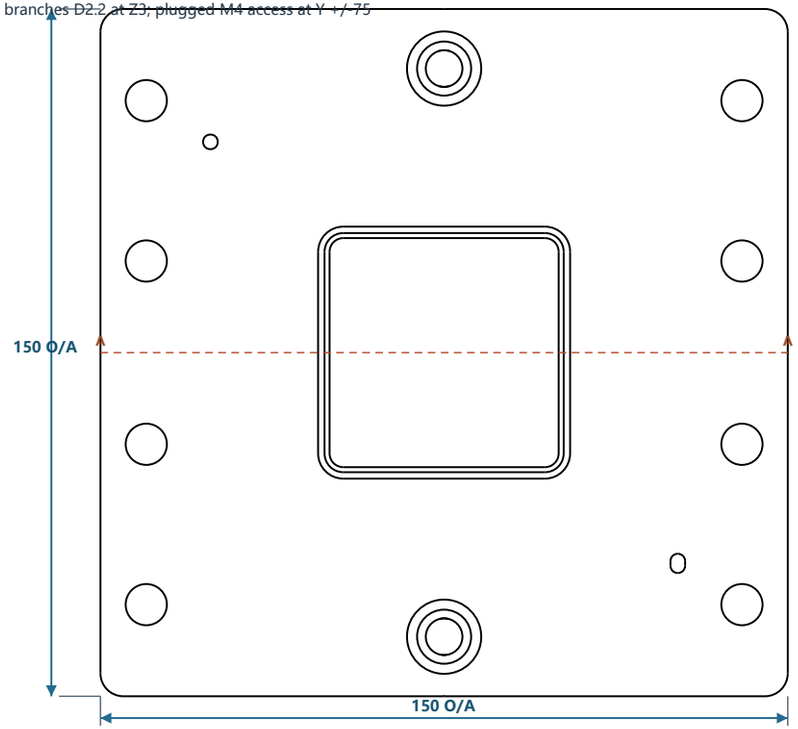
ASSEMBLY	DRAWING	SOURCE	STATUS
7-layer compact plate-press	**KOLBE-7 / ASM-7**	Kolbe_7.FCStd Stack.Shape	**REVIEW**

KF-101 | 流道框架

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 150 x 150 x 6; R5 frame
02 50 x 50 wet window, R3; no headers
03 Main seal groove: 1.4 wide x 0.75 deep each face
04 2 x D8 manifolds at (0, +/-62); mean D14 seal groove
05 Buried branches D2.2 at Z3; plugged M4 access at Y +/-75



PLAN / FRONT (1:1.2)

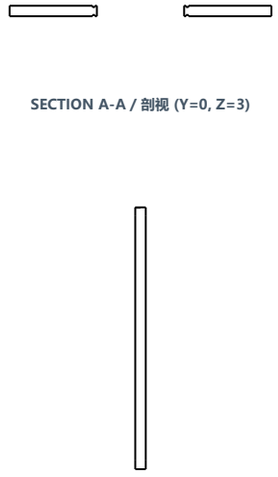
MATERIAL / 材料

PVDF provisional; chemistry and pressure rating pending

TOLERANCES / 公差

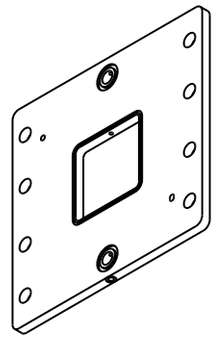
Unspecified unless individually stated; confirm before release.

PVDF provisional; chemistry/pressure and sealing qualification pending



SECTION A-A / 剖视 (Y=0, Z=3)

SIDE / 侧视 (1:3.1)



ISOMETRIC / 轴测 (1:3.1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KF-101 / DWG-KF-101**	See each view; dimensions govern	**REVIEW**

KP-101 | A向正极箔片

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 56 x 56 x 0.35 coupon; R2 corners
02 A-side dry tab to +X; width 16; tip X=100
03 D4.5 tab hole at X=90; no manifold or locator holes
04 Wet-window / coupon footprint ratio: 79.56%; tab excluded

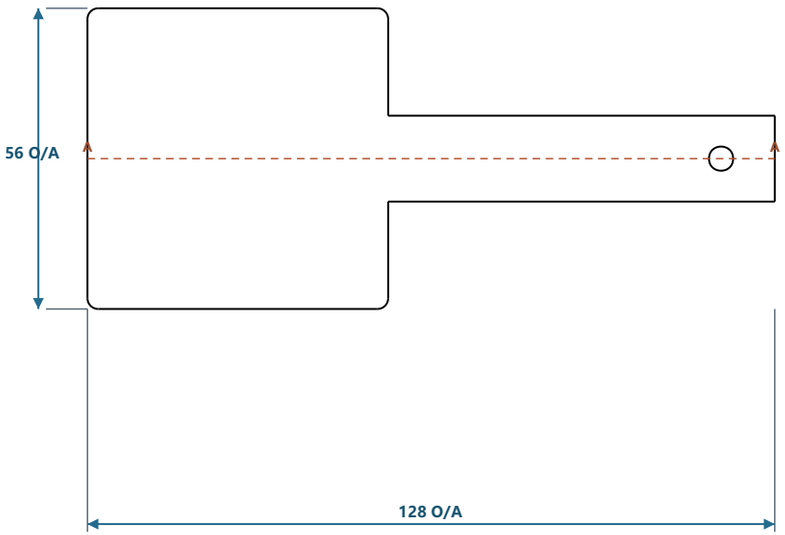
MATERIAL / 材料

Platinum or Pt-Ti; final specification TBD

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

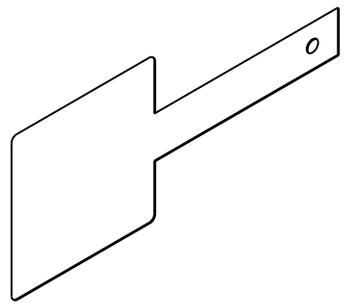
Platinum or Pt-Ti; final specification TBD



PLAN / FRONT (1:1.0)

SECTION A-A / 剖视 (Y=0, Z=0.175)

SIDE / 侧视 (1:1.2)



ISOMETRIC / 轴测 (1:1.5)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-101 / DWG-KP-101**	See each view; dimensions govern	**REVIEW**

KP-102 | C向负极箔片

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 56 x 56 x 0.35 coupon; R2 corners
02 C-side dry tab to -X; width 16; tip X=-100
03 D4.5 tab hole at X=-90; no manifold or locator holes
04 Wet-window / coupon footprint ratio: 79.56%; tab excluded

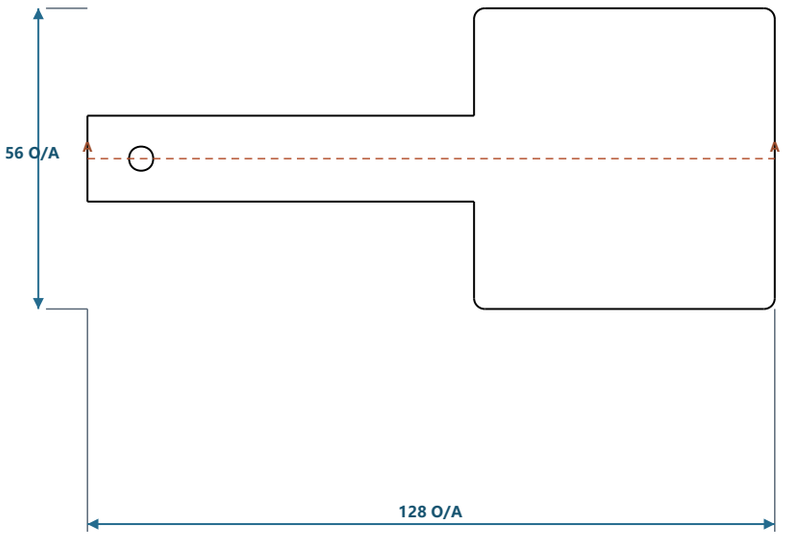
MATERIAL / 材料

Material TBD

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

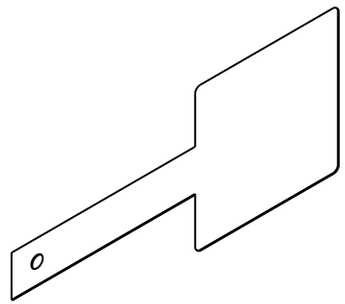
Material TBD



PLAN / FRONT (1:1.0)

SECTION A-A / 剖视 (Y=0, Z=0.175)

SIDE / 侧视 (1:1.2)



ISOMETRIC / 轴测 (1:1.5)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

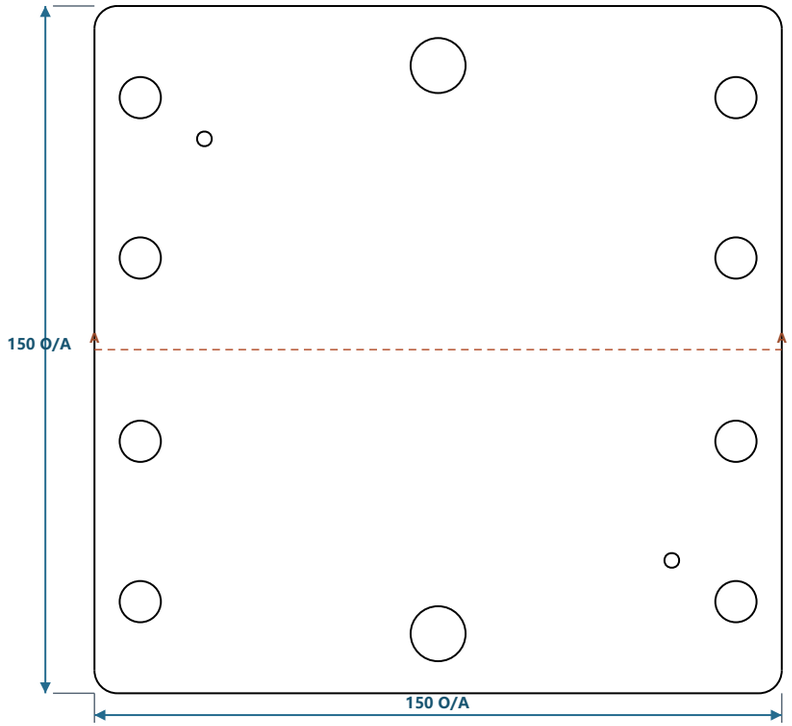
ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-102 / DWG-KP-102**	See each view; dimensions govern	**REVIEW**

KP-201 | 前端板

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 150 x 150 x 12 endplate; R5
02 8 x D9 through bolt clearance
03 2 x D12 tool-access through holes at (0, +/-62)
04 2 x D3.2 x 3 deep inner locator blind holes



PLAN / FRONT (1:1.2)

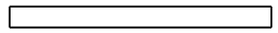
MATERIAL / 材料

Aluminum alloy TBD; coating and compatibility TBD

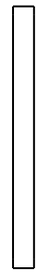
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

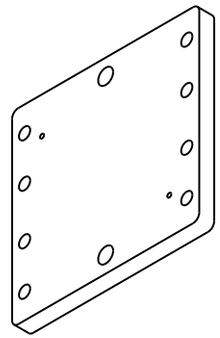
Aluminum alloy TBD; coating and compatibility TBD



SECTION A-A / 剖视 (Y=0, Z=6)



SIDE / 侧视 (1:3.1)



ISOMETRIC / 轴测 (1:3.1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

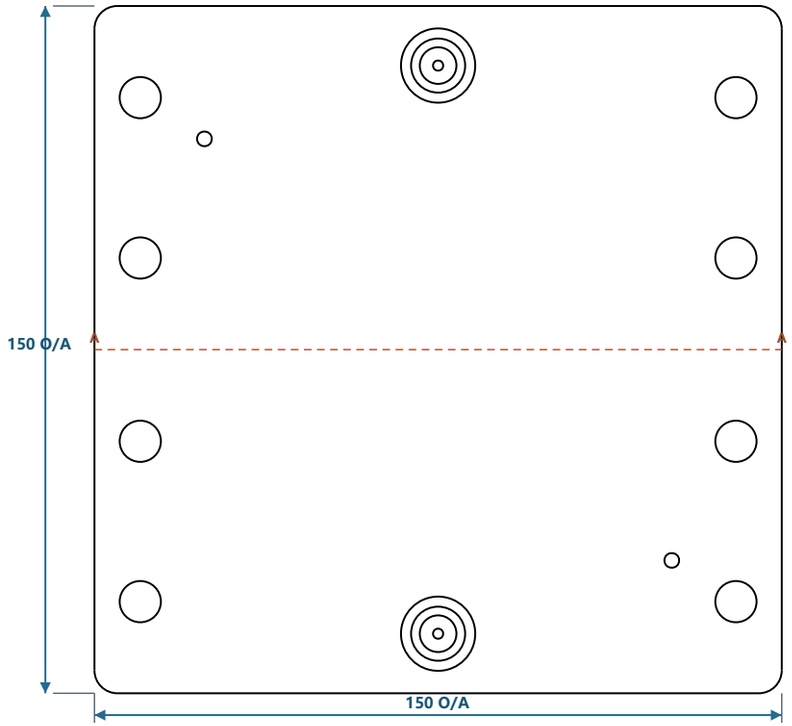
ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KP-201 / DWG-KP-201	See each view; dimensions govern	REVIEW

KP-202 | 前绝缘板

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 150 x 150 x 14 front insulator; R5
02 8 x D9 bolts + 2 x D3.2 through locators
03 6 front ports at X=+/-32.5, +/-19.5, +/-6.5; M6x1 nominal envelope
04 Port stack: D6 x 8; D3.4 seat x 2; D2.2 pilot x 1; inner grooves mean D14 x 2.2W x 1.1 deep



PLAN / FRONT (1:1.2)

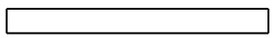
MATERIAL / 材料

PEEK, unfilled; material grade TBD

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

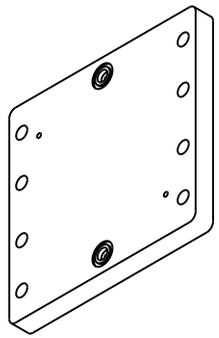
PEEK, unfilled; material grade TBD



SECTION A-A / 剖视 (Y=0, Z=7)



SIDE / 侧视 (1:3.1)



ISOMETRIC / 轴测 (1:3.1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-202 / DWG-KP-202**	See each view; dimensions govern	**REVIEW**

KP-203 | 后绝缘板

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 150 x 150 x 6 rear insulator; R5
02 8 x D9 bolt holes + 2 x D3.2 through locator holes
03 Inner face (Z0): 2 x D8 x 2 blind manifolds
04 2 x annular grooves mean D14, 2.2 wide x 1.1 deep

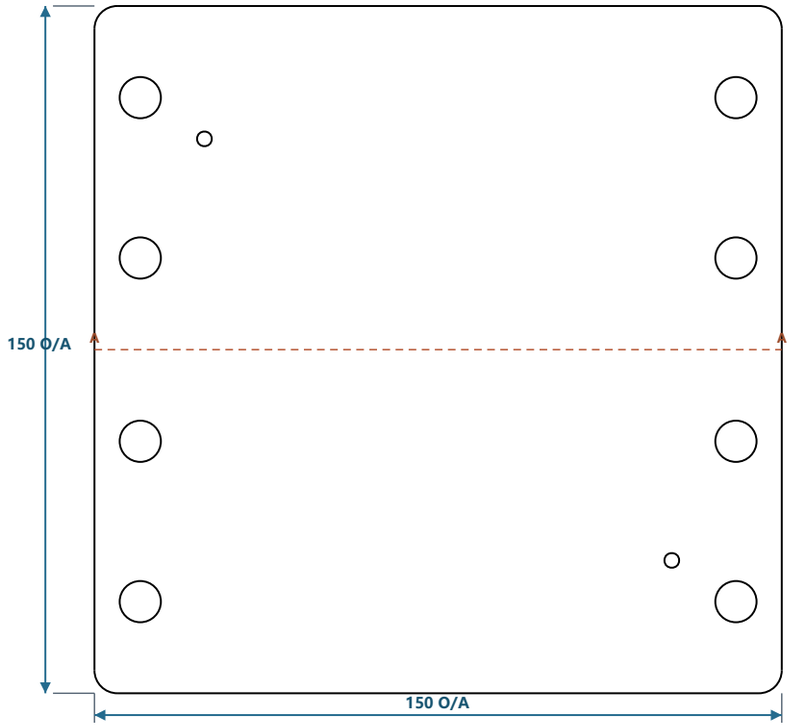
MATERIAL / 材料

PEEK, unfilled; material grade TBD

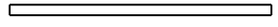
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

PEEK, unfilled; material grade TBD



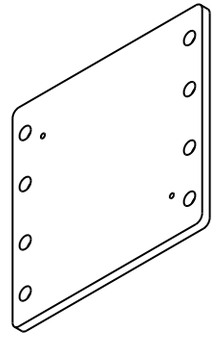
PLAN / FRONT (1:1.2)



SECTION A-A / 剖视 (Y=0, Z=3)



SIDE / 侧视 (1:3.1)



ISOMETRIC / 轴测 (1:3.1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

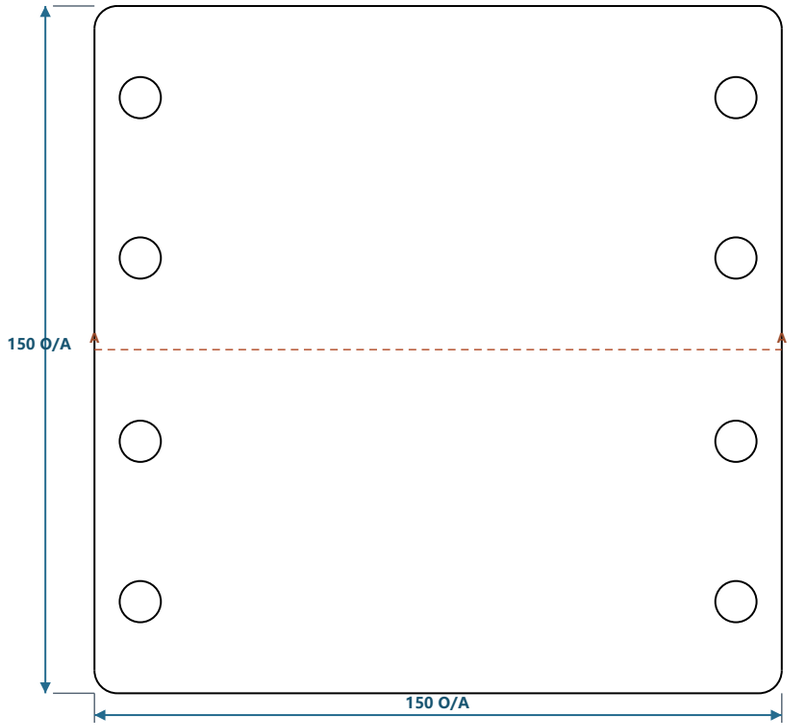
ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-203 / DWG-KP-203**	See each view; dimensions govern	**REVIEW**

KP-204 | 后端板

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 150 x 150 x 12 rear endplate; R5
02 8 x D9 through bolt clearance
03 No fluid-access holes
04 2 x D3.2 x 3 deep inner locator blind holes



PLAN / FRONT (1:1.2)

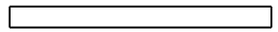
MATERIAL / 材料

Aluminum alloy TBD; coating and compatibility TBD

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

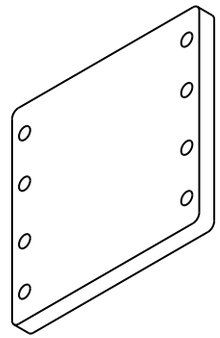
Aluminum alloy TBD; coating and compatibility TBD



SECTION A-A / 剖视 (Y=0, Z=6)



SIDE / 侧视 (1:3.1)



ISOMETRIC / 轴测 (1:3.1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

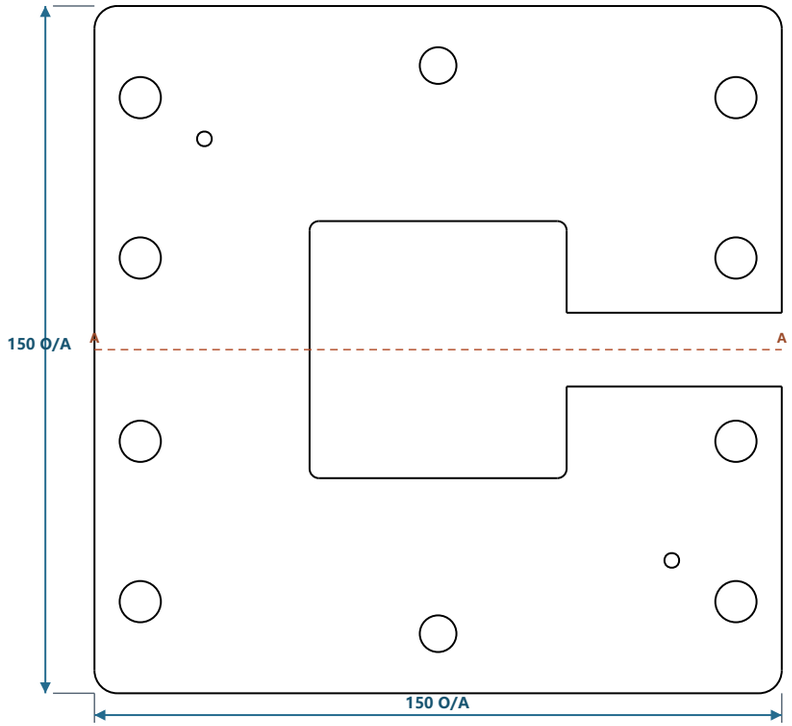
ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-204 / DWG-KP-204**	See each view; dimensions govern	**REVIEW**

KP-601 | A侧电极载体膜

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 150 x 150 x 0.35 PEEK carrier film
02 56.10 aperture, R2.05; terminal slot W16.10 to +X
03 2 x D8 manifold holes at (0, +/-62)
04 8 x D9 bolts; 2 x D3.2 locator holes; thickness tolerance provisional +/-0.01



PLAN / FRONT (1:1.2)

MATERIAL / 材料

PEEK precision film; supplier and grade TBD

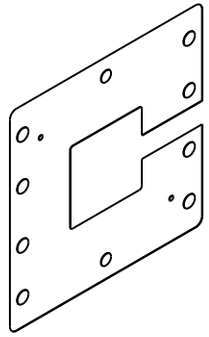
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

PEEK precision film; supplier and grade TBD

SECTION A-A / 剖视 (Y=0, Z=0.175)

SIDE / 侧视 (1:3.1)



ISOMETRIC / 轴测 (1:3.0)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

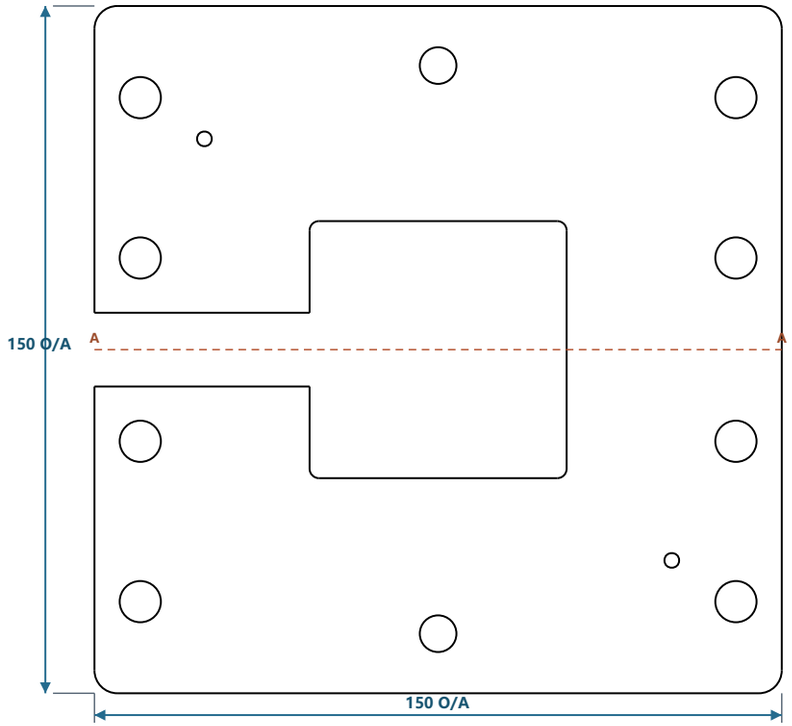
ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KP-601 / DWG-KP-601	See each view; dimensions govern	REVIEW

KP-602 | C侧电极载体膜

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 150 x 150 x 0.35 PEEK carrier film
02 56.10 aperture, R2.05; terminal slot W16.10 to -X
03 2 x D8 manifold holes at (0, +/-62)
04 8 x D9 bolts; 2 x D3.2 locator holes; thickness tolerance provisional +/-0.01



PLAN / FRONT (1:1.2)

MATERIAL / 材料

PEEK precision film; supplier and grade TBD

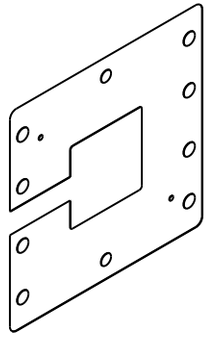
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

PEEK precision film; supplier and grade TBD

SECTION A-A / 剖视 (Y=0, Z=0.175)

SIDE / 侧视 (1:3.1)



ISOMETRIC / 轴测 (1:3.0)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KP-602 / DWG-KP-602	See each view; dimensions govern	REVIEW

KP-301 | 夹紧爪

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 16 x 16 x 3 clamp jaw; R1

02 D4.5 through hole

03 Material: PEEK unfilled (provisional)

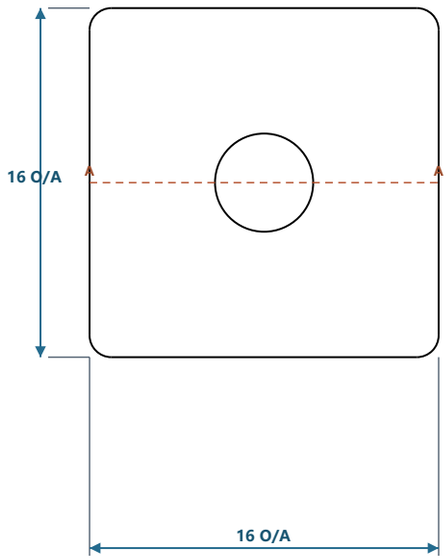
MATERIAL / 材料

PEEK, unfilled; material grade TBD

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

PEEK, unfilled; material grade TBD



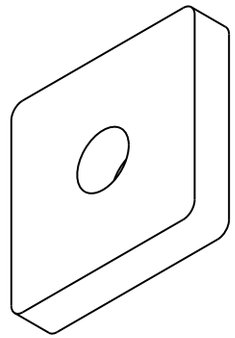
PLAN / FRONT (4.0:1)



SECTION A-A / 剖视 (Y=0, Z=1.5)



SIDE / 侧视 (3.0:1)



ISOMETRIC / 轴测 (3.0:1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-301 / DWG-KP-301**	See each view; dimensions govern	**REVIEW**

KP-501 | 流体接头

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 M6x1 nominal smooth envelope, D6 x 8 long

02 Head D9 x 7; bore D3.1 through

03 Thread is not modeled helically; 316L stainless

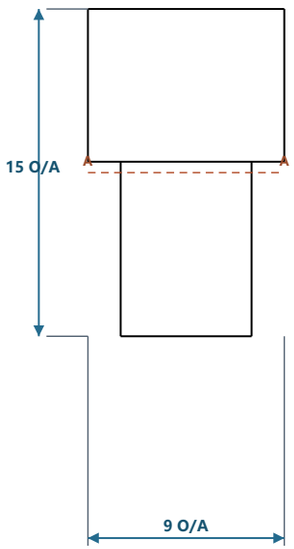
MATERIAL / 材料

316L stainless steel; M6x1 thread represented by nominal envelope

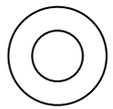
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

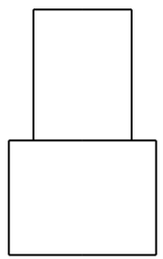
316L stainless steel; M6x1 thread represented by nominal envelope



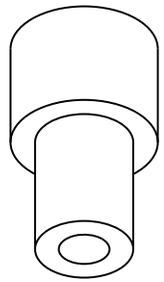
PLAN / FRONT (4.0:1)



SECTION A-A / 剖视 (Y=0, Z=0)



SIDE / 侧视 (3.0:1)



ISOMETRIC / 轴测 (3.0:1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-501 / DWG-KP-501**	See each view; dimensions govern	**REVIEW**

KP-502 | 压缩密封圈

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Compression seal: OD3.4 x 2 long

02 Modeled bore D3.0; free-state bore D2.95

03 PTFE supplier/fit TBD

MATERIAL / 材料

PTFE, unfilled; supplier TBD

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

PTFE, unfilled; supplier TBD



SECTION A-A / 剖视 (Y=0, Z=0)



PLAN / FRONT (4.0:1)



SIDE / 侧视 (3.0:1)



ISOMETRIC / 轴测 (3.0:1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-502 / DWG-KP-502**	See each view; dimensions govern	**REVIEW**

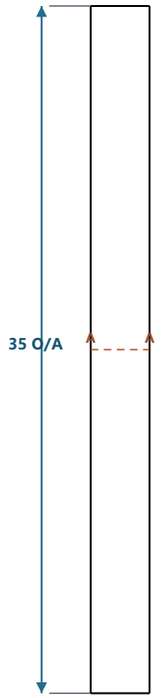
KP-503 | PTFE管

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Tube OD3 / ID1.6 x 35 long

02 PTFE; chemistry and pressure suitability TBD



PLAN / FRONT (3.6:1)

MATERIAL / 材料

PTFE, unfilled; pressure and chemical suitability TBD

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

PTFE, unfilled; pressure and chemical suitability TBD



SECTION A-A / 剖视 (Y=0, Z=0)



SIDE / 侧视 (1.4:1)



ISOMETRIC / 轴测 (2.0:1)

GAP 6 mm IS A REVIEW ENVELOPE ONLY; FINAL ELECTRODE/GAP/FLOW ARE TBD. No pressure rating or final torque assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KP-503 / DWG-KP-503**	See each view; dimensions govern	**REVIEW**

KB-A3 | A向三层铜梳排

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 3-layer A positive-X copper comb; 6 x 8 spine section
02 Finger indices i=1; active height 13.05 mm; layer pitch 6.35
03 Finger blank 34 x 8 x 2; through-hole Ø4.5 along Z at X=90
04 2 x M4x0.7 tapped cross-holes, core Ø3.3, through the 6 mm spine along X at Z=1 and H-3

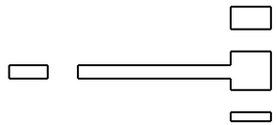
MATERIAL / 材料

Copper, grade TBD; current rating not established

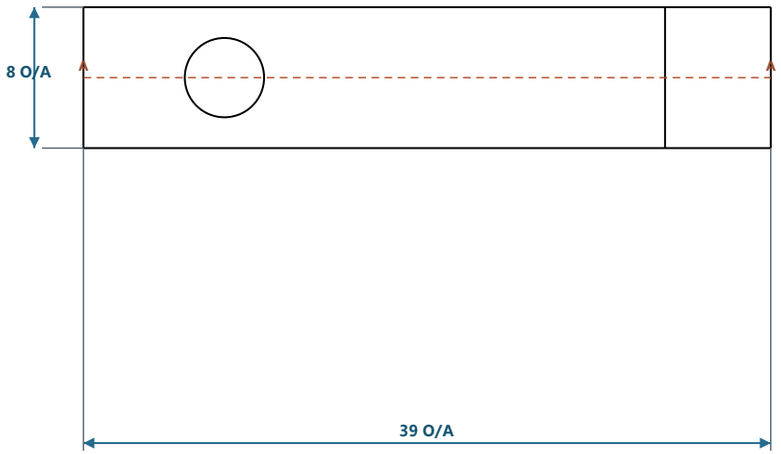
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

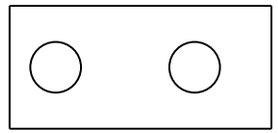
Copper grade TBD; allowable current rating not established; beam cross-section preliminary



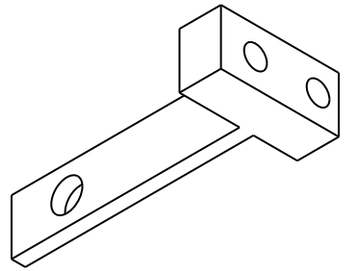
SECTION A-A / 剖视 (Y=0, Z=6.525)



PLAN / FRONT (3.2:1)



SIDE / 侧视 (2.8:1)



ISOMETRIC / 轴测 (1.8:1)

Copper grade/current rating unconfirmed; finger beam section is preliminary. No current capacity assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KB-A3 / DWG-KB-A3	See each view; dimensions govern	REVIEW

KB-C3 | C向三层铜梳排

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 3-layer C negative-X copper comb; 6 x 8 spine section
02 Finger indices i=0,2; active height 13.05 mm; layer pitch 6.35
03 Finger blank 34 x 8 x 2; through-hole Ø4.5 along Z at X=-90
04 2 x M4x0.7 tapped cross-holes, core Ø3.3, through the 6 mm spine along X at Z=1 and H-3

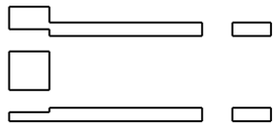
MATERIAL / 材料

Copper, grade TBD; current rating not established

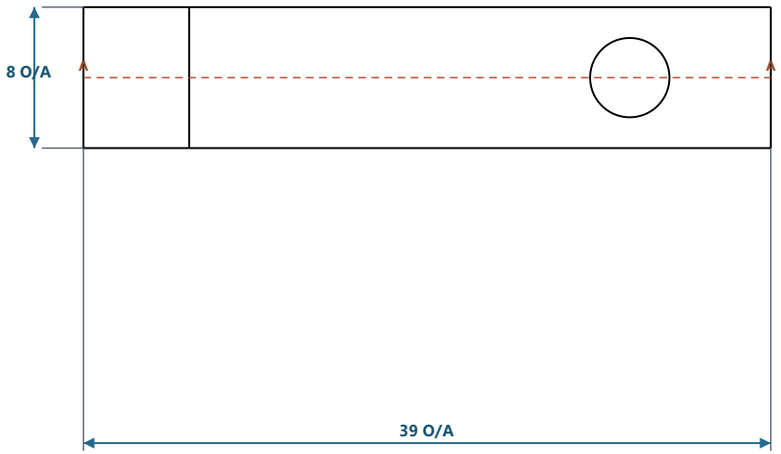
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

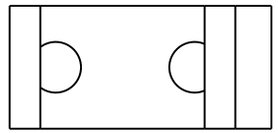
Copper grade TBD; allowable current rating not established; beam cross-section preliminary



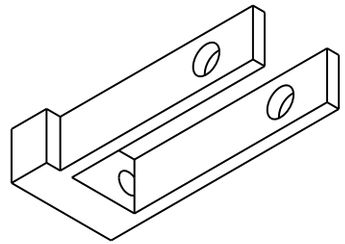
SECTION A-A / 剖视 (Y=0, Z=6.525)



PLAN / FRONT (3.2:1)



SIDE / 侧视 (2.8:1)



ISOMETRIC / 轴测 (1.5:1)

Copper grade/current rating unconfirmed; finger beam section is preliminary. No current capacity assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KB-C3 / DWG-KB-C3	See each view; dimensions govern	REVIEW

KB-A5 | A向五层铜梳排

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 5-layer A positive-X copper comb; 6 x 8 spine section
02 Finger indices i=1,3; active height 25.75 mm; layer pitch 6.35
03 Finger blank 34 x 8 x 2; through-hole Ø4.5 along Z at X=90
04 2 x M4x0.7 tapped cross-holes, core Ø3.3, through the 6 mm spine along X at Z=1 and H-3

MATERIAL / 材料

Copper, grade TBD; current rating not established

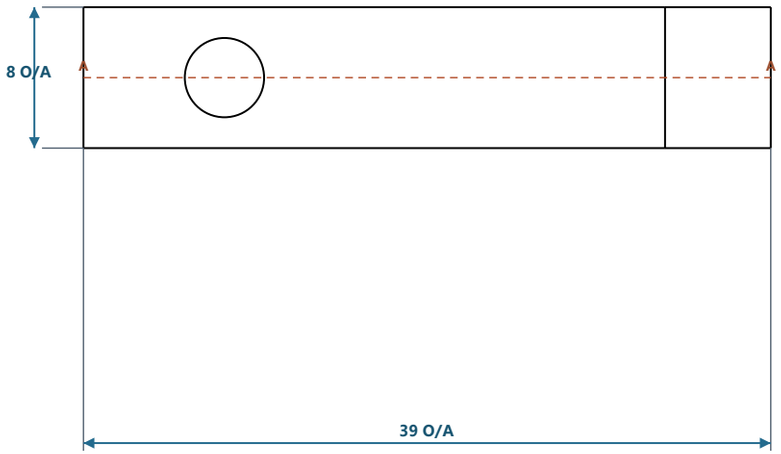
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

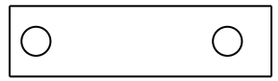
Copper grade TBD; allowable current rating not established; beam cross-section preliminary



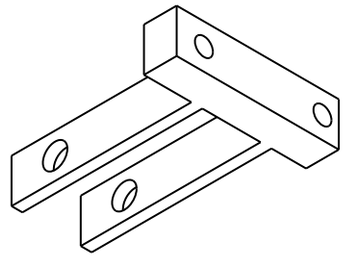
SECTION A-A / 剖视 (Y=0, Z=12.8751)



PLAN / FRONT (3.2:1)



SIDE / 侧视 (1.6:1)



ISOMETRIC / 轴测 (1.4:1)

Copper grade/current rating unconfirmed; finger beam section is preliminary. No current capacity assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KB-A5 / DWG-KB-A5**	See each view; dimensions govern	**REVIEW**

KB-C5 | C向五层铜梳排

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 5-layer C negative-X copper comb; 6 x 8 spine section
02 Finger indices i=0,2,4; active height 25.75 mm; layer pitch 6.35
03 Finger blank 34 x 8 x 2; through-hole Ø4.5 along Z at X=-90
04 2 x M4x0.7 tapped cross-holes, core Ø3.3, through the 6 mm spine along X at Z=1 and H-3

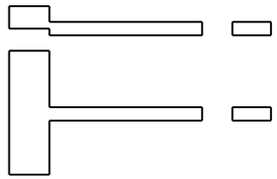
MATERIAL / 材料

Copper, grade TBD; current rating not established

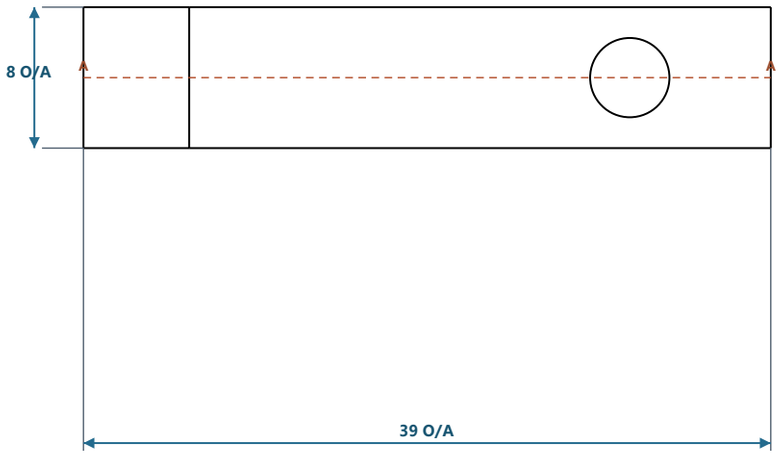
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

Copper grade TBD; allowable current rating not established; beam cross-section preliminary



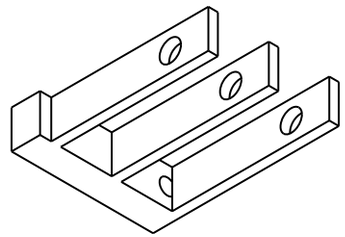
SECTION A-A / 剖视 (Y=0.2=12.875)



PLAN / FRONT (3.2:1)



SIDE / 侧视 (1.6:1)



ISOMETRIC / 轴测 (1.2:1)

Copper grade/current rating unconfirmed; finger beam section is preliminary. No current capacity assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KB-C5 / DWG-KB-C5**	See each view; dimensions govern	**REVIEW**

KB-A7 | A向七层铜梳排

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 7-layer A positive-X copper comb; 6 x 8 spine section
02 Finger indices i=1,3,5; active height 38.45 mm; layer pitch 6.35
03 Finger blank 34 x 8 x 2; through-hole Ø4.5 along Z at X=90
04 2 x M4x0.7 tapped cross-holes, core Ø3.3, through the 6 mm spine along X at Z=1 and H-3

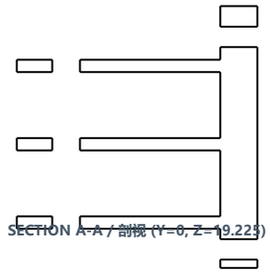
MATERIAL / 材料

Copper, grade TBD; current rating not established

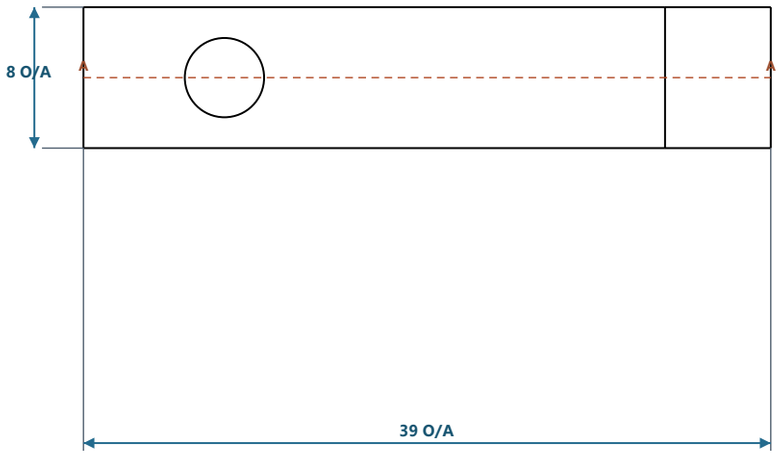
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

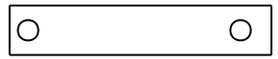
Copper grade TBD; allowable current rating not established; beam cross-section preliminary



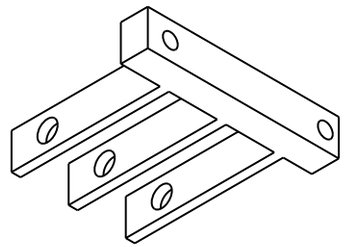
SECTION A-A / 剖视 (Y=0, Z=19.225)



PLAN / FRONT (3.2:1)



SIDE / 侧视 (1.1:1)



ISOMETRIC / 轴测 (1.2:1)

Copper grade/current rating unconfirmed; finger beam section is preliminary. No current capacity assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KB-A7 / DWG-KB-A7	See each view; dimensions govern	REVIEW

KB-C7 | C向七层铜梳排

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 7-layer C negative-X copper comb; 6 x 8 spine section
02 Finger indices i=0,2,4,6; active height 38.45 mm; layer pitch 6.35
03 Finger blank 34 x 8 x 2; through-hole Ø4.5 along Z at X=-90
04 2 x M4x0.7 tapped cross-holes, core Ø3.3, through the 6 mm spine along X at Z=1 and H-3

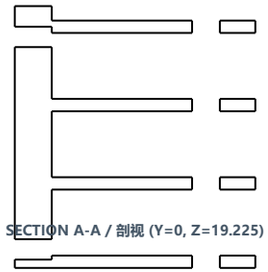
MATERIAL / 材料

Copper, grade TBD; current rating not established

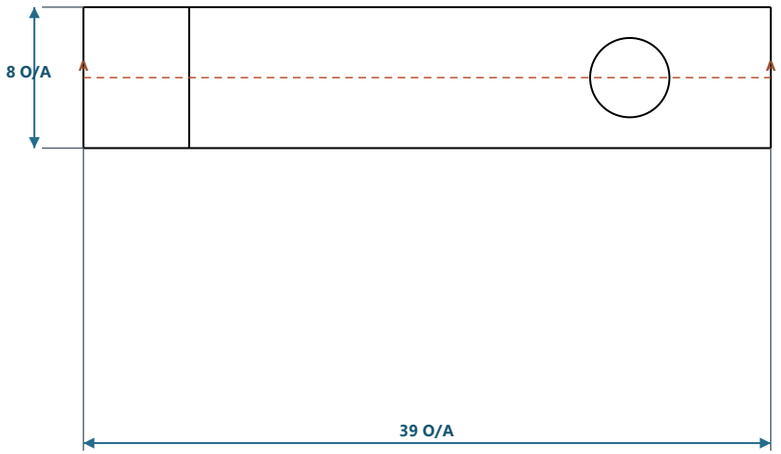
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

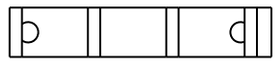
Copper grade TBD; allowable current rating not established; beam cross-section preliminary



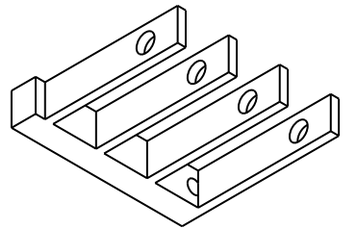
SECTION A-A / 剖视 (Y=0, Z=19.225)



PLAN / FRONT (3.2:1)



SIDE / 侧视 (1.1:1)



ISOMETRIC / 轴测 (1.0:1)

Copper grade/current rating unconfirmed; finger beam section is preliminary. No current capacity assigned.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KB-C7 / DWG-KB-C7	See each view; dimensions govern	REVIEW

KB-801 | 前PEEK支架

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Foot 70 x 60 x 5; 2 x Ø9 tie-rod clearance through at (65, +/-20)
02 Post section 7 x 12; Z=-0.5..31.5; transverse comb hole Ø4.5 at Y=0, Z=27
03 Fixing holes align with existing M8 tie-rod stack; no separate clamp/strap fastener
04 Front bracket in native coordinates; align A/C comb support cross-hole

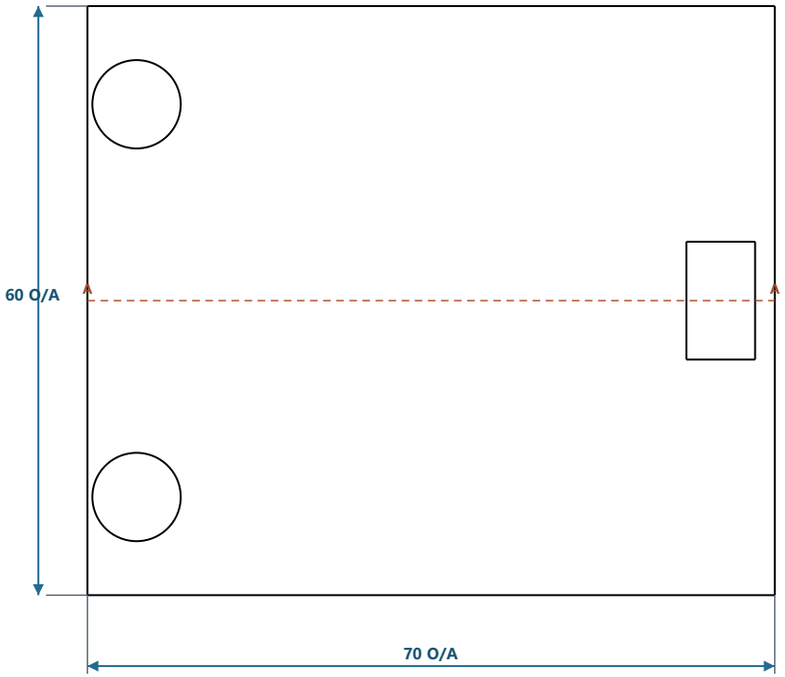
MATERIAL / 材料

PEEK, grade TBD; structural qualification pending

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

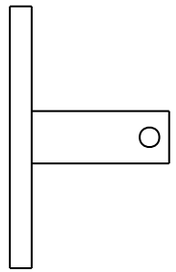
PEEK grade TBD; preliminary cantilever; structural qualification pending



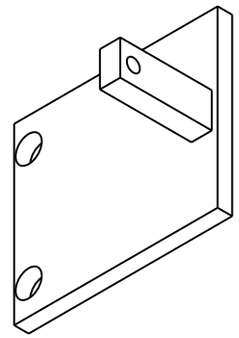
PLAN / FRONT (1.8:1)



SECTION A-A / 剖视 (Y=0, Z=13.25)



SIDE / 侧视 (1:1.2)



ISOMETRIC / 轴测 (1:1.3)

PEEK support is a preliminary cantilever; structural qualification pending.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KB-801 / DWG-KB-801**	See each view; dimensions govern	**REVIEW**

KB-802 | 后PEEK支架

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Foot 70 x 60 x 5; 2 x Ø9 tie-rod clearance through at (65, +/-20)

02 Post section 7 x 12; Z=-0.5..25; transverse comb hole Ø4.5 at Y=0, Z=21

03 Fixing holes align with existing M8 tie-rod stack; no separate clamp/strap fastener

04 Rear bracket assembly transform: flip 180 deg about X, then translate to stack height

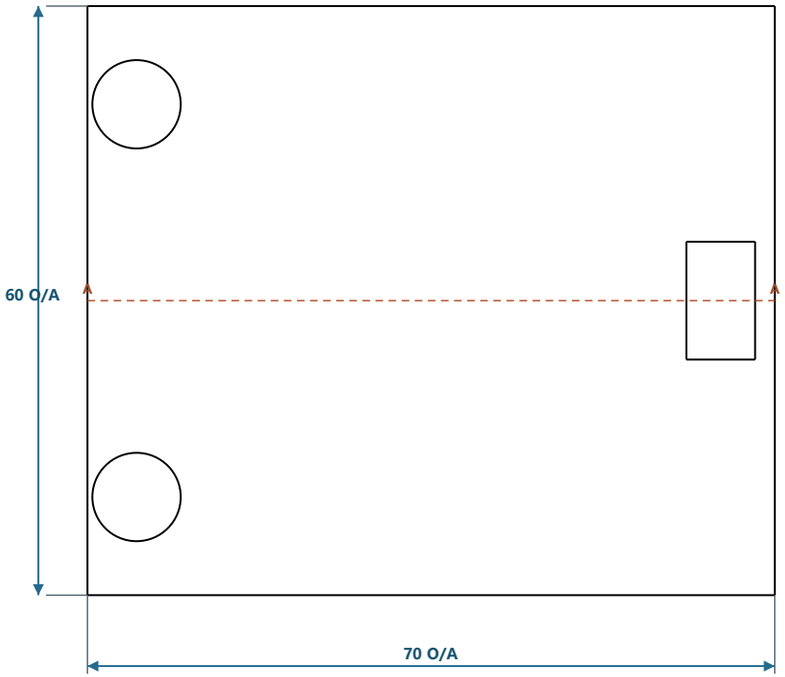
MATERIAL / 材料

PEEK, grade TBD; structural qualification pending

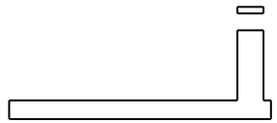
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

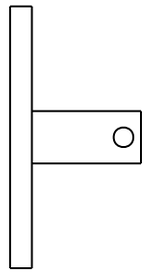
PEEK grade TBD; preliminary cantilever; structural qualification pending



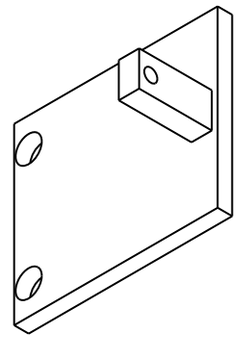
PLAN / FRONT (1.8:1)



SECTION A-A / 剖视 (Y=0, Z=10)



SIDE / 侧视 (1:1.2)



ISOMETRIC / 轴测 (1:1.3)

PEEK support is a preliminary cantilever; structural qualification pending.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KB-802 / DWG-KB-802**	See each view; dimensions govern	**REVIEW**

KH-901 | M4交叉孔堵头

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Axis global +Y; M4x0.7 nominal smooth Ø3.3 envelope x 6 long
02 Low-profile head Ø5.5 x 2; driver slot 1.0 wide x 0.8 deep
03 Thread is a smooth nominal envelope; helical thread is not modeled
04 Fit/seal compatibility with plugged manufacturing access requires qualification

MATERIAL / 材料

316L stainless steel; exact grade and thread fit TBD

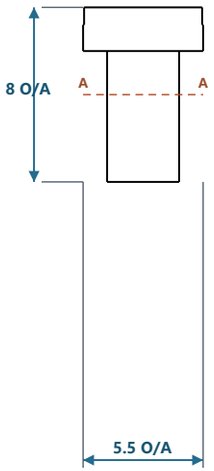
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

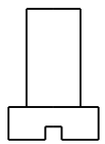
316L exact grade and thread fit TBD; no vendor certification implied



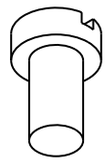
SECTION A-A / 剖视 (Y=0, Z=0)



PLAN / FRONT (4.0:1)



SIDE / 侧视 (3.0:1)



ISOMETRIC / 轴测 (3.0:1)

Thread/washer fit and plug sealing require qualification; thread is not modeled helically.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KH-901 / DWG-KH-901	See each view; dimensions govern	REVIEW

KH-902 | 堵头垫圈

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Washer OD5.5 / ID4.2 x 0.5 thick
02 Used beneath KH-901 plug head
03 Stock/material and sealing fit TBD

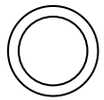
MATERIAL / 材料

Washer stock TBD

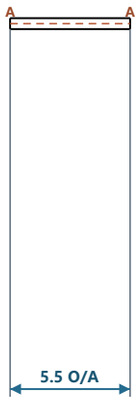
TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

Washer stock and sealing fit TBD

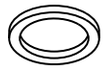


SECTION A-A / 剖视 (Y=0, Z=0)



PLAN / FRONT (4.0:1)

SIDE / 侧视 (3.0:1)



ISOMETRIC / 轴测 (3.0:1)

Thread/washer fit and plug sealing require qualification; thread is not modeled helically.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	KH-902 / DWG-KH-902	See each view; dimensions govern	REVIEW

KS-101 | 主圆角方形密封圈

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Closed rounded-square seal centerline 53.6 x 53.6; centerline R4.8

02 Free cross-section diameter Ø1.0 (diameter, not radius)

03 Centerline path corresponds to the KF-101 main groove; installed squeeze not assigned

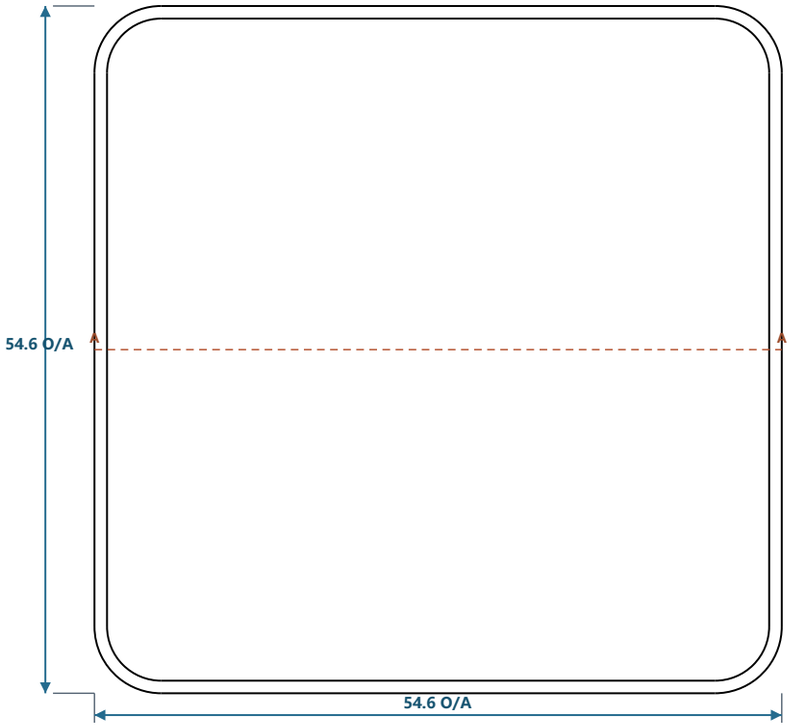
MATERIAL / 材料

Elastomer TBD; seal compound and hardness unresolved

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

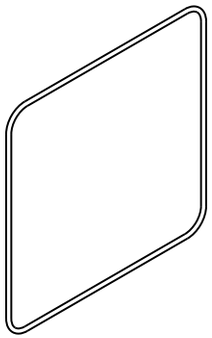
Elastomer compound, hardness, compression and chemical compatibility TBD



PLAN / FRONT (2.3:1)

SECTION A-A / 剖视 (Y=0, Z=0)

SIDE / 侧视 (1:1.1)



ISOMETRIC / 轴测 (1:1.1)

Elastomer compound, hardness, compression and chemical compatibility remain TBD.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KS-101 / DWG-KS-101**	See each view; dimensions govern	**REVIEW**

KS-102 | 歧管圆形密封圈

BREP-derived orthographic projections | First-angle orientation | Not for manufacture

FEATURE / DIMENSION SCHEDULE | 特征尺寸表

01 Circular seal mean/centerline diameter Ø14.0

02 Free cross-section diameter Ø1.5 (diameter, not radius)

03 For paired axial manifold seal grooves; installed squeeze not assigned

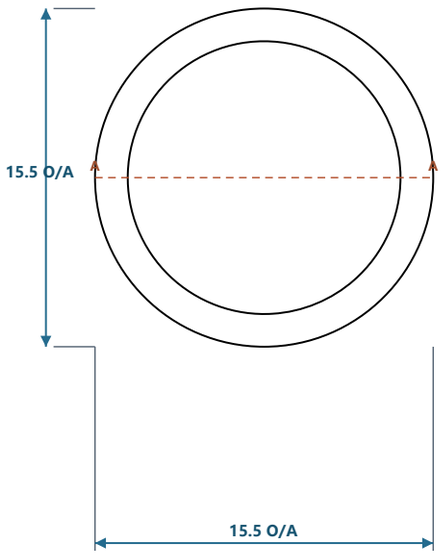
MATERIAL / 材料

Elastomer TBD; seal compound and hardness unresolved

TOLERANCES / 公差

Unspecified unless individually stated; confirm before release.

Elastomer compound, hardness, compression and chemical compatibility TBD



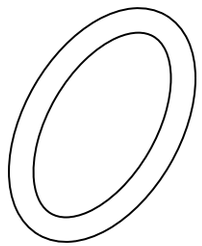
PLAN / FRONT (4.0:1)



SECTION A-A / 剖视 (Y=0, Z=0)



SIDE / 侧视 (3.0:1)



ISOMETRIC / 轴测 (3.0:1)

Elastomer compound, hardness, compression and chemical compatibility remain TBD.

ASSEMBLY	PART / DRAWING	SCALE / 比例	STATUS
Compact plate-press cell	**KS-102 / DWG-KS-102**	See each view; dimensions govern	**REVIEW**